

# Chapter 1

## TT

### 1.1 SYS-0

The alphabet is listed as follows.

- ◇  $V = \{v_i\}_{i \geq 0} \cup \{A_i\}_{i \geq 0}$ ;
- ◇  $C$  is the set of constant symbols;
- ◇  $B = \{\Pi, \lambda, \Sigma, \dots\}$  is the set of binders;
- ◇  $U = \{\text{Type}_i\}_{i \geq 0}$  is the set of universes;
- ◇  $P = \{(\cdot), \cdot, \cdot, \equiv\}$  is the set of punctuation.

All of these sets are at most countable.

The whole formulas we use is contained by set

$$F = V|C|U|B(V : F)F|(FF)|(CF^*).$$

**Notation 1.1.1** *Here we announce some commonly used notations for avoiding ambiguity.*

- ◇  $\mathbf{a} = a_0 a_1 \dots a_m$ ;
- ◇  $f(\mathbf{x}) := f$  where  $f \in F$  and  $\text{Fr } f \subset \{x_0, \dots, x_m\}$ ;
- ◇  $(f\mathbf{x})$  refers  $(\dots((fx_0)x_1)\dots)$ , note that this may differ the notation  $(fx_0x_1 \dots x_m)$ , which is equal to our first definition;
- ◇ for any  $b \in B$ ,  $b(\mathbf{v} : \mathbf{A})N := b(v_0 : A_0)b(v_1 : A_1) \dots b(v_m : A_m)N$ ;
- ◇  $\mathbf{v} : \mathbf{A}$  refers
 
$$v_0 : A_0 \quad v_1 : A_1 \quad \dots \quad v_s : A_s;$$
- ◇  $M \rightarrow N := \Pi(v : M)N$ , in which  $v \notin \text{Fr } N$ ;
- ◇ to distinguish between terms in  $F$  and constants in  $C$ , we use a mono font for constants;
- ◇  $\mathcal{U} := \text{Type}_{\aleph_0}$ , and  $M : \mathcal{U}$  implies there exists a  $k < \aleph_0$  such that  $M : \text{Type}_k$ .
- ◇ for any relation  $\mathcal{R} \subset F^2$ ,  $\text{Compatible}\mathcal{R}$  is a abbreviation of the statement: for all if  $MRN$ , then for any  $L, A, \phi_1, \dots, \phi_s \in F$ ,  $b \in B$ ,  $\mathbf{c} \in C$  and  $v \in V$ ,  $(ML)\mathcal{R}(NL)$ ,  $(LM)\mathcal{R}(LN)$ ,  $b(v : A)MRb(v : A)N$  and

$$(\mathbf{c}\phi_1 \dots \phi_k M \phi_{k+1} \dots \phi_s)\mathcal{R}(\mathbf{c}\phi_1 \dots \phi_k N \phi_{k+1} \dots \phi_s)$$

**Definition 1.1.2** A definition  $D$  has the form

$$\mathbf{c}(v : A) \equiv d(v) : B,$$

where  $v \in V$ ,  $\mathbf{c} \in C$  and all of  $A, B, d(v)$  are in  $F$ .

**Definition 1.1.3** A sequent has the form

$$D \quad v : A \quad M : A$$

where  $M : A$  is called a judgement, and both  $M$  and  $A$  are in  $F$ . The set of all sequents is denoted by  $\text{Sqt}$ .

**Legal judgements** consist a subset of  $F : F$ , which is derived from a series of **rules**. In our SYS-0, we first introduce the following rules, each of them can be viewed as a relation on  $\text{Sqt}^* \times \text{Sqt}$ .

### Sort

If  $m < n$ , then

$$\_ \_ \text{Type}_m : \text{Type}_n$$

### Var

If  $v \in V \setminus (\Gamma \cap V)$ , then

$$\begin{array}{l} D \quad \Gamma \quad A : \mathcal{U} \\ D \quad \Gamma \quad v : A \quad v : A \end{array}$$

### Weak

If  $v \in V \setminus (\Gamma \cap V)$ , then

$$\begin{array}{l} D \quad \Gamma \quad M : A \\ D \quad \Gamma \quad B : \mathcal{U} \\ D \quad \Gamma \quad v : B \quad M : A \end{array}$$

## 1.2 SYS-1

我们通过引入新符号, 增加新规则和关系来增加新的项和类型, 这个过程大致遵循一个固定的工作流: 1. Formation: 引入新规则, 让系统在指定条件下可以导出新类型; 2. Constuction: 引入新规则, 让系统在指定条件下可以导出新项, 并且这些项都具有新类型; 3. Elimination: 引入新规则, 告知系统新项应该如何使用; 4. Computaiton: 引入  $F$  上的新关系, 告知系统新系统中某些表达和旧系统中的某些表达具有同样的语义.

现在我们将拓展 SYS-0, 增加  $\lambda$ -Terms 和  $\Pi$ -Types.

### SYS-1: Form

If  $s_1 \leq s_2$ , then

$$\begin{array}{l} D \quad \Gamma \quad A : \mathcal{U} \\ D \quad \Gamma \quad v : A \quad B : \mathcal{U} \\ D \quad \Gamma \quad \Pi(v : A)B : \mathcal{U} \end{array}$$

### SYS-1: Constr

$$\begin{array}{l} D \quad \Gamma \quad v : A \quad M : B \\ D \quad \Gamma \quad \Pi(v : A)B : \mathcal{U} \\ D \quad \Gamma \quad \lambda(v : A)M : \Pi(v : A)B \end{array}$$

**SYS-1: Elim**

$$\begin{array}{l}
\mathbf{D} \quad \Gamma \quad M : \Pi(v : A)B \\
\mathbf{D} \quad \Gamma \quad N : A \\
\mathbf{D} \quad \Gamma \quad (MN) : B \frac{N}{v}
\end{array}$$

**SYS-1: Compu**

**Definition 1.2.1**  $\rightarrow_1 \subset F^2$  is the samllest relation satisfies the following properties:

- ◇ for any  $A, M, N$  in  $F$ ,  $(\lambda(v : A)MN) \rightarrow_1 M \frac{N}{v}$ ;
- ◇ Compatible  $\rightarrow_1$ .

Based on this newly defiened relation, we introduce the Computaiton Rule of SYS-1 as follows. If  $B =_1 B'$ , then

$$\begin{array}{l}
\mathbf{D} \quad \Gamma \quad M : B \\
\mathbf{D} \quad \Gamma \quad B' : \mathcal{U} \\
\mathbf{D} \quad \Gamma \quad M : B'
\end{array}$$

**1.3 SYS-2**

在 SYS-1 的基础上加入  $\Sigma$ -Types 得到 SYS-2.

**SYS-2: Form**

$$\begin{array}{l}
\mathbf{D} \quad \Gamma \quad A : \mathcal{U} \\
\mathbf{D} \quad \Gamma \quad v : A \quad B : \mathcal{U} \\
\mathbf{D} \quad \Gamma \quad \Sigma(v : A)B
\end{array}$$

**SYS-2: Constr**

$$\begin{array}{l}
\mathbf{D} \quad \Gamma \quad \Sigma(v : A)B \\
\mathbf{D} \quad \Gamma \quad a : A \\
\mathbf{D} \quad \Gamma \quad b : B \frac{a}{v} \\
\mathbf{D} \quad \Gamma \quad (\text{pair}ab) : \Sigma(v : A)B
\end{array}$$

定义 Elim 前先明确我需求, 对任何 Type 或者 Universe  $Q$ , 通过给定的项  $\phi : \Pi(v : A)(Bv \rightarrow Q)$ , 其中  $Bv$  指的是公式的合并  $(Bv)$ , 我们可以构造出一个具有类型  $(\Sigma(v : A)Bv) \rightarrow Q$  的项  $\psi$ . 所以, 有理由将规则设计成下面这个样子:

$$\begin{array}{l}
\mathbf{D} \quad \Gamma \quad \Sigma(v : A)B : \mathcal{U} \\
\mathbf{D} \quad \Gamma \quad \phi : \Pi(v : A)(Bv \rightarrow Q) \\
\mathbf{D} \quad \Gamma \quad (\mathbf{Elim}\phi Q) : (\Sigma(v : A)Bv) \rightarrow Q
\end{array}$$

然后再加入新关系  $\rightarrow_2$ , 其包含下面内容

$$((\mathbf{Elim}\phi Q)(\text{pair}ab)) \rightarrow_2 ((\phi a)b)$$

除此之外还有一种更普遍的方法, 它是从更复杂的功能中下放到这里的.

**SYS-2: Elim**

$$\begin{array}{l}
D \quad \Gamma \quad \Sigma(v : A)B : \mathcal{U} \\
D \quad \Gamma \quad (\mathbf{Ind}\Sigma(v : A)B) : \Pi(X : (\Sigma(v_0 : A)B) \rightarrow s). \\
\quad [\Pi(v_1 : A)(v_2 : Bv_2)(X\mathbf{pair}_{v_1v_2}) \rightarrow \Pi(v_3 : \Sigma(v_4 : A)Bv_4)(Xv_3)]
\end{array}$$

可见如果给出常函数  $H = \lambda(X : \Sigma(v_0 : A)B)Q : \Pi(X : (\Sigma(v_0 : A)B) \rightarrow s)$ , 还有

$$\phi : \Pi(v_0 : A)(v_1 : Bv_0)H(\mathbf{pair}_{v_0v_1}) = \Pi(v_0 : A)(v_1 : Bv_0)Q,$$

那么

$$(\mathbf{Ind}\Sigma(v : A)B)(H\phi) : (\Sigma(v : A)Bv) \rightarrow Q.$$

**SYS-2: Compu**

**Definition 1.3.1**  $\rightarrow_2 \subset F^2$  is the samllest relation satisfies the following properties:

$$\begin{array}{l}
\Diamond \text{ for all } a, b, A, B, \phi, H \in F, \\
\quad (\mathbf{Ind}\Sigma(v : A)B)(H\phi ab) \rightarrow_2 (\phi a)b
\end{array}$$

$$\Diamond \text{ Compatible } \rightarrow_2.$$

$=_2$  is the equivalence genertated by  $\rightarrow_1 \cup \rightarrow_2$ .

**1.4 SYS-3**

在 SYS-2 中加入自然数类型  $\mathbb{N}$  使其成为 SYS-3.

**SYS-3: Form**

$$\begin{array}{l}
D \quad \Gamma \quad \bar{\mathbb{N}} : \mathcal{U} \\
D \quad \Gamma \quad \mathbb{N} : \mathcal{U}
\end{array}$$

**SYS-3: Constr**

$$\begin{array}{l}
D \quad \Gamma \quad \mathbb{N} : \mathcal{U} \\
D \quad \Gamma \quad \mathbf{0} : \mathbb{N}
\end{array}$$

and

$$\begin{array}{l}
D \quad \Gamma \quad \mathbb{N} : \mathcal{U} \\
D \quad \Gamma \quad \mathbf{S} : \mathbb{N} \rightarrow \mathbb{N}
\end{array}$$